

A Dual-Tank Model for Real-Time Tracking of Phosphocreatine and Glycolytic Energy Reserves in Cycling

White paper — AnaerobicFuelTanks project Version 0.7 · 2026-07-12 (recovery-law fix: the linear LT1 gate is replaced by Skiba's intensity-dependent recovery shape, so the model now completes a standard 4×4; the tanks are reframed as compartments of W', not of muscle metabolites — §4.1a; §6.9's mechanism is corrected to the glycolytic flux ceiling; and a new §6.10 runs the model against biopsy data and reports where it fails)

Scope and status (read this once). This is a **physiologically-motivated decomposition of one measured quantity** — single-tank W'bal — whose split fraction **f_p** is assumed, not measured (power cannot identify the depletion split; §4.2). The two bars are **compartments of W'**, named for their dominant metabolic system; the metabolic identification is *motivation, not a measured claim* — W' recovers ~4× faster than the muscle metabolites it is named after (§4.1a), so the "PCr bar" is a fast-reserve-of-W' readout, not a muscle-PCr gauge. The characterisation is narrow: **during steady supra-CP effort both bars are affine in W'bal** ($R_p/C_p \approx R_g/C_g \approx W'_{bal} + \text{const}$; §4.2, §4.4). But the earlier claim that the PCr bar is affine "most of the time" is **withdrawn**: under the oxidative-headroom recovery gate the effective PCr recovery constant is 100–500 s while a rider is actually pedalling (not 27 s), so the PCr bar is *live* for a substantial fraction of interval-ride time (§6.8 measures 20–79%). The "affine most of the time" caveat applies to stopped or easy recovery, not to riding. **Non-redundant content lives in the transients**: the glycolytic *activation ramp* at onset (τ_{on} measured, τ_{off} not) and the *PCr recovery transient* after hard efforts. **There is still no experiment showing the two-bar decomposition is more correct than single-tank W'bal** — that requires the recovery-law head-to-head (§6.6). The v0.6 claim that the compartments are "unfalsifiable in-modality — permanently" is also **withdrawn** (§6.10): muscle biopsy during cycle ergometry is in-modality compartment-level data, it is already in the bibliography, and when the model is run against it (repeated-sprint ATP partitioning) it **fails in level and direction** — the first genuinely falsifiable result in the project. What survives: v0.5–0.6's **usefulness** arguments. (i) On real interval rides the PCr bar is informative ~45% of the time (§6.8). (ii) For **training**, the cumulative per-system load separates an alactic from a glycolytic session — but the separation is driven by the **glycolytic flux ceiling scaled by the ramp** (not the ramp alone, as v0.6 claimed; §6.9), the metric is blind to *submaximal* alactic work, and it must beat a zero-parameter "fraction of work in efforts <15 s" stopwatch baseline, not just W'bal. **f_p**, **τ_{off}** , **g_rate**, the emergent **τ_{dep}** , and both recovery gates (**gate_p** and the LT1-band shape) are assumed. The rest of the paper builds on these; it does not re-litigate them.

Abstract

Endurance-cycling analytics have converged on the Critical Power (CP) / W' framework, in which all work performed above a sustainable power asymptote is drawn from a single finite reserve, W' . This lumps two physiologically distinct anaerobic systems — the fast phosphocreatine (PCr / alactic) system and the slower glycolytic (lactic) system — into one tank with a single recovery time constant. That simplification is convenient but discards the information a rider most wants during hard, variable-intensity efforts: *which* reserve is being spent, and *how fast* each will come back. This paper proposes a **dual-tank model** that splits W' into a fast PCr reserve and a slow glycolytic reserve, drives both from the instantaneous power trace, and applies system-specific depletion and restoration laws drawn from the bioenergetic and ^{31}P -MRS literature. We are explicit up front about what this is: a **physiologically-motivated decomposition of one measured quantity (W' bal) whose split is assumed, not measured**, and whose extra resolution over single-tank W' bal is real but concentrated in the transients after hard efforts. It generalizes W' bal in depletion and diverges from it — by design — in recovery. The model is deliberately reduced to be computable at 1 Hz on a wrist- or bar-mounted device, and we specify a Garmin Connect IQ implementation that displays live reserve, consumption, and depletion for both systems. We close with a validation strategy — including the one out-of-sample test that would actually earn the second tank — and an explicit statement of the model's assumptions and limits.

1. Introduction

A power meter reports mechanical output, not metabolic cost. The bridge between the two is a model of the body's energy-supply systems. The dominant bridge in cycling — the CP/ W' model — treats supra-CP work as depletion of a single reservoir and sub-CP work as its exponential refill (Skiba et al. 2012). It is power-native, validated, and computationally trivial, which is why it underlies W' bal displays in GoldenCheetah, intervals.icu, and several head units.

Its central simplification is also its central weakness for real-time decision-making: **W' is one tank**. Physiologically, anaerobic capacity is at least two tanks with very different dynamics (Margaria; di Prampero & Ferretti 1999):

- The **phosphocreatine (PCr / alactic)** system — high power, low capacity, **fast** oxidative recovery (seconds to a minute), but with recovery that is pH-sensitive and slows across repeated bouts.
- The **glycolytic (lactic)** system — larger capacity, lower peak power, and **slow** recovery (minutes) that effectively only proceeds at low intensity.

A rider deciding whether to cover the next surge cares which tank is empty. If the PCr tank is full but glycolytic is depleted, a 5-second jump is fine but a 40-second dig is not — a distinction the single-tank W' bal cannot express. The goal of this paper is a model that keeps the power-native, on-device virtues of W' bal while restoring the two-system resolution.

There are two distinct use cases, and they are informed by different outputs. The first — *pacing/racing* — uses the live bars in the moment ("can I cover this attack?"); its value is concentrated in the post-effort transients (§7), and is real but narrow. The second — **training prescription and load monitoring** — uses the *cumulative per-system load* over a session, and it is where the decomposition is most clearly worth more than W'_{bal} (§6.9). The two anaerobic systems carry very different **physiological and recovery costs**: PCr-based (alactic) surges — short, high-power, with adequate recovery — resynthesise in seconds-to-a-minute and can be repeated at high volume with little systemic fatigue or metabolite accumulation; glycolytic surges accumulate H^+/Pi and clear over many minutes (lactate $t_{1/2} \approx 1366$ s; Ferguson 2010), and cost far more recovery. An athlete who wants to *target* the PCr system — alactic power, repeat-sprint ability — needs to keep efforts short and recoveries full enough that glycolysis stays disengaged, and to know, per session, how much of the anaerobic work was alactic vs glycolytic. W'_{bal} collapses that to one number; the two-tank model does not (§6.9).

Two honest qualifications, stated here rather than buried in §7. First, the split is a **modeling assumption**, not a measurement — power alone cannot identify how W' divides between the two systems (§4.2). Second, the extra resolution is **concentrated in transients**: whenever the fast reserve is full the glycolytic reserve is just an affine rescaling of single W'_{bal} . But note the fast reserve is *not* full "most of the time" while riding: under the oxidative-headroom recovery gate (§4.2) its effective recovery constant is 100–500 s at typical soft-pedal powers, not the 27 s of a stopped rider, so the bar stays live through the recovery valleys of an interval session (§6.8: 20–79% of ride time). The model earns its keep as a *heuristic decomposition of W'* , not as two independently measured reserves — and, per §4.1a, the tanks are compartments of W' , not of muscle metabolites.

2. Physiological background

Phosphocreatine (alactic). ATP is buffered by the creatine-kinase reaction ($PCr + ADP \rightarrow ATP + Cr$). Intramuscular PCr ($\sim 20\text{--}25$ mmol $\cdot$ kg $^{-1}$ wet muscle) is the highest-power energy source and is drawn on instantly at any power transient. Its resynthesis is oxidative and fast: classic alactic O_2 -debt half-time ≈ 30 s (Margaria); ^{31}P -MRS gives recovery time constants $\tau_{PCr} \approx 20\text{--}40$ s, strongly **pH-dependent** (acidosis slows it) and **biphasic** under high glycolytic load (Harris et al. 1976; Yoshida et al. 2013, PMID 23662804). Recovery also **slows progressively across repeated hard bouts**.

Glycolytic (lactic). Anaerobic glycolysis converts glycogen to lactate, supplying large ATP flux for ~ 10 s–2 min. Its "fuel level" is best tracked as accumulated muscle/blood lactate — a proxy for H^+ and metabolite accumulation. Critically, **glycolytic flux itself falls across repeated maximal sprints**: acidosis inhibits phosphorylase and phosphofructokinase, so the glycolytic ATP contribution decays from $\approx 40\%$ of total ATP in the first of 10×6 s efforts to $<10\%$ by the last (Gaitanos 1993; the biopsy time-courses reviewed in *Sci Rep* 2024, DOI 10.1038/s41598-024-78916-z), while the **aerobic** contribution rises to absorb the shortfall ($\sim 29\% \rightarrow 43\%$ between sprints; Bogdanis 1996). Restoration of the glycolytic reserve is slow (lactic O_2 -debt half-time ≈ 15 min) and, mechanistically, proceeds fastest at low intensity, effectively ceasing above the first

lactate threshold (LT1). *The model as shipped has no term for glycolytic flux falling with fatigue — a documented failure it now runs against explicitly (§6.10), with an optional term to partially address it.*

The onset that matters. Both systems contribute from the start of a hard effort, but not equally at first: PCr is drawn on instantly (it is the highest-power, immediate buffer), while glycolytic flux takes several seconds to ramp up as glycogen phosphorylase is activated (Parolin et al. 1999) — so the *early* demand is PCr-weighted, shifting toward a shared draw as glycolysis engages. This graded onset — and the very different refill rates — is what the dual-tank model must reproduce (it does so with a parallel, PCr-weighted draw and a glycolytic activation ramp; §4.2), not a strict "PCr first, then glycolytic" hand-off.

What the two "tanks" represent — and what they don't. The "tank" is a deliberate simplification for intuition and on-device display, not a claim that either system is a literal reservoir that drains. The two sides are not even the same *kind* of quantity:

- The **PCr tank** does correspond to a real, depletable substrate store — intramuscular phosphocreatine — so "PCr empty" is close to literal.
- The **glycolytic "tank"** is better understood as **tolerance to accumulating fatigue-related metabolites** than as a fuel gauge. On the ~1–4 min timescale it represents, muscle glycogen is nowhere near exhausted; carbohydrate depletion is a separate, hours-long limiter this model does not address. What forces power down at the limit of a hard effort is the buildup of inorganic phosphate (Pi), H⁺ (acidosis), ADP, and extracellular K⁺ toward the limit of tolerance — consistent with exhaustion at CP/W' coinciding with a reproducible low-PCr / high-metabolite state. (Lactate itself is largely a fuel and a marker, not the cause of force loss.)

This is the physiological reason the two recovery laws differ: PCr resynthesizes quickly, whereas the glycolytic side recovers slowly and only at low intensity because it reflects *clearance and buffering* of accumulated byproducts, which need aerobic metabolism and time — exactly why the model tracks the glycolytic state as an accumulation variable (a muscle-lactate proxy; §4.1) and gates its recovery below LT1. Muscle fatigue is multifactorial and still debated; W' deliberately lumps it into one number for usability. In short: **the PCr tank is a fuel that depletes; the glycolytic "tank" is a byproduct bucket that fills.**

3. Prior models (and why a reduced dual-tank sits between them)

Three model families exist (full survey in `literature-review-anaerobic-models.md`):

- **Family A — CP / W'-balance** (Skiba 2012/2015; Bartram 2018; Froncioni–Clarke). Power-native and on-device-friendly, but **one lumped tank, single recovery τ** . Recovery $\tau_{W'} = 546 \cdot e^{(-0.01 \cdot D_{CP})} + 316 \text{ s}$ cannot represent fast PCr + slow glycolytic simultaneously.
- **Family B — hydraulic tank models** (Morton 1986; Weigend `three_comp_hyd` 2021). **Two explicit anaerobic vessels** (AnF = phosphagen, AnS = glycolytic) plus an aerobic vessel; out-predicts W'bal

for intermittent recovery (arXiv 2108.04510). But its 8 parameters are abstract and fit per-athlete by evolutionary computation — heavy for an embedded target.

- **Family C — bioenergetic supply/demand ODEs** (EJAP 2023, PMID 37369795). Explicit alactic / lactic / aerobic metabolic-rate terms, with an efficiency-limited PCr recovery and a power-gated lactate-removal law. Mechanistically ideal but has 17 parameters and needs grey-box fitting.

The gap: Family A is implementable but under-resolved; Families B and C resolve the two systems but are too heavy to run and calibrate on a head unit. The dual-tank model below keeps the two-tank resolution of B/C with the on-device simplicity of A.

To be precise about what "lighter" means here, because it is easy to oversell: the model's core free-parameter count (CP , W' , f_p , P_{p_max} , g_{rate} , τ_p , τ_g , τ_{on} , $LT1$, with $\tau_{off} = \tau_{on}$ by default — **nine free numeric parameters**, plus *three* optional realism terms and *two* functional-form gates — $gate_p$ (PCr oxidative-headroom) and the intensity-dependent glycolytic recovery shape — each an assumption in its own right, now flagged in §7) is essentially the hydraulic model's eight — **the same size, not a smaller model**. The difference is *where the parameters come from*. Only CP and W' are fitted per athlete (from a standard CP test); the rest are **hard-coded to literature defaults** rather than fit by evolutionary computation. That buys on-device simplicity and a trivial calibration, at a real cost: the load-bearing defaults (f_p , P_{p_max} , $LT1$, and the recovery τ 's) are exactly the quantities a CP test does **not** determine, so the model's two-system resolution rests on assumptions, not measurements. That is a legitimate identifiability-for-convenience trade — but it is a trade, and the sections below are explicit about which numbers are earned and which are assumed.

4. The dual-tank model

4.1 State and parameters

Two energy reserves (Joules), each with a fixed capacity:

- R_p — **PCr reserve**, capacity C_p
- R_g — **glycolytic reserve**, capacity C_g
- Total anaerobic capacity $W' = C_p + C_g$

The model has **nine free numeric parameters** (plus $\tau_{off} = \tau_{on}$, three optional realism terms, and two functional-form gates — see §3). Only CP and W' come from a CP test; the rest are literature-fixed defaults (see §3 — this is a real count, not a small one):

Symbol	Meaning	Typical default
CP	critical power (W)	from 3-/12-min CP test ($\approx$ FTP + a few %)
W'	total work above CP (J)	from same test ($\sim$ 15–25 kJ)

Symbol	Meaning	Typical default
f_p	fraction of W' assigned to the PCr tank	0.25 — <i>assumed</i> (physiology, not data; §7)
P_{p_max}	PCr peak power above CP (W), at a full tank	$\approx P_{1s} - CP$ (best 1 s power – CP)
g_rate	glycolytic peak flux as a fraction of PCr peak flux (a ratio)	0.5 — load-bearing (di Prampero & Ferretti 1999); band ~ 0.3 – 0.7 , drives §6.9's headline (sep 48 $\rightarrow$ 33 pts across the band); flag, not a settled constant
τ_p	fast-reserve recovery time constant (s) — a W' -recovery constant, <i>not</i> muscle-PCr resynthesis	27 — set by the W' -reconstitution curve (§4.1a), <i>exonerated</i> there; do not read the bar as biopsy PCr (W' recovers $\sim 4\times$ faster; §4.1a). Gated by oxidative headroom (§4.2)
τ_g	slow-reserve recovery time constant (s)	470 — reconstitution fit (Ferguson 2010 at 20 W $\Rightarrow$ gate ≈ 0.90 ; joint optimum with f_p ; §4.1a)
τ_{on}	glycolytic activation time constant (s)	6 (Parolin 1999)
τ_{off}	glycolytic de -activation time constant (s)	= τ_{on} — <i>assumed</i> , and now shown not load-bearing (a $20\times$ change moves §6.9 by 1–2 pts; §6.9, §7)
LT1	first lactate threshold power (W) — sets the <i>band</i> of the glycolytic recovery shape	measured (a threshold test), <i>not</i> a fixed %CP
$gate_p$	PCr oxidative-headroom recovery gate $(CP - P)/CP$ — a <i>functional form</i> , no free constant	assumed (v0.5); shape is a choice (linear/ $\sqrt{\hspace{-0.05em}}\hspace{-0.05em}$ /sigmoid swing the headline recovery number ~ 21 – 31 pts; §7)
g_fat	<i>optional</i> glycolytic flux-fatigue exponent, $rate_g \cdot (R_g/C_g)^{\{g_fat\}}$	0 (off) by default — moves repeated-sprint partition toward the biopsy direction but does not reproduce it (§6.10)

Derived: $C_p = f_p \cdot W'$, $C_g = (1 - f_p) \cdot W'$. Initial state $R_p = C_p$, $R_g = C_g$ (full).

A note on naming (new in v0.7). We call the two reserves the **fast reserve** (R_p , "PCr") and the **slow reserve** (R_g , "glycolytic"). The metabolic labels are **motivation, not a measured claim**: §4.1a shows the reserves are compartments of W' , with the right internal fast/slow ratio ($\sim 18\times$) but an absolute timescale $\sim 4\times$ faster than the muscle metabolites they are named after — because W' itself recovers $\sim 4\times$ faster than PCr or lactate. Every "the bar equals muscle PCr" reading in earlier drafts is therefore withdrawn; the bar is a fast-recovering component of W' bal. This rename is the honest resolution of the identifiability problem the review series has circled (Reviewer 2, third report): the model is right about W' 's internal dynamics and agnostic about the metabolites.

Free-parameter count. Nine free numeric parameters (CP , W' , f_p , P_{p_max} , g_rate , τ_p , τ_g , τ_{on} , LT1); $\tau_{off} = \tau_{on}$ by default. Two functional-form gates ($gate_p$, the glycolytic recovery

shape) and three optional realism terms (pH-slowed PCr recovery, aerobic ramp, glycolytic flux fatigue) sit on top. Only **CP** and **W'** are fitted per athlete; the rest are literature-set. That is essentially the hydraulic model's eight — **the same size, but literature-set rather than fitted** (the real trade; §3). A fast- reserve depletion kinetic, $\tau_{\text{dep}} = C_p / P_{p_max}$, also falls out of the above (§4.4) and is equally assumed — see §7.

η was removed (was 0.80). The former "recovery efficiency" only rescaled the effective PCr recovery constant to τ_p / η — no sub- C_p asymptote, so not a true efficiency, and degenerate with τ_p when fitted. It silently made the effective constant ≈ 27.6 s while the table advertised 22 s. **$\tau_p = 27$ s is now justified on its own terms:** it sits inside the 31P-MRS τ_{PCr} range of 20–40 s (Yoshida 2013; Harris 1976) — *not* as "22 inflated by η " (that would just re-import the double-count). The **eta** settings key remains, defaulting to 1.0 (identity) with a **deprecation note**, so a future maintainer cannot silently reintroduce the rescale. (The pulmonary-VO₂ recovery half-time of 74 s in Ferguson 2010 is *slower* than muscle τ_{PCr} , because it includes circulatory processes — see the component-mismatch discussion in §4.1a.)

The $g_{pmax} = 0.5 \cdot P_{p_max}$ ratio is not arbitrary: peak glycolytic ATP flux is roughly half peak PCr/creatine-kinase flux in human muscle (di Prampero & Ferretti 1999; the biopsy time-courses of Parolin 1999 / Bogdanis 1996), so PCr is the higher-power system by about 2:1. **But g_rate is now a load-bearing parameter, not an aside:** the ablation in §6.9 shows it (via the glycolytic flux ceiling) is what generates the training-load separation, and sweeping it 0.3 → 0.7 moves that headline separation from ~48 to ~33 pts. It is flagged alongside **f_p** and given a band (~0.3–0.7), and it should be sanity-checked against sprint biopsy data (§6.10), which it has never been fit to.

Several of these deserve flags up front, because the defaults are doing real work:

- **f_p is assumed on physiological grounds, but now corroborated by the recovery data (§4.1a).** We set **0.25** from physiological alactic-fraction estimates (the PCr system supplies ~20–30% of anaerobic ATP capacity; di Prampero & Ferretti 1999, Bangsbo 1990), and — once the reconstitution reference protocol's recovery power is known (20 W, near-passive) — the reconstitution curve independently implies $f_p \approx 0.20\text{--}0.25$, converging with the physiology. Treat it as a per-athlete calibration target; the supported band is ~0.20–0.25 (a joint reconstitution fit centres on 0.20).
- **$P_{p_max} \approx P_{1s} - CP$ is the rate at a FULL tank.** PCr flux is not constant: the rate *ceiling* tapers with tank fullness (creatine-kinase equilibrium — flux falls as PCr depletes), so the available PCr rate is $P_{p_max} \cdot (R_p / C_p)$, equal to P_{p_max} only when the tank is full (§4.2). Read **$P_{1s} - CP$** as an upper bound: at 1 s glycolysis is already ~15% active, so it mildly over-attributes to PCr. Note this taper is on the *ceiling* only — it does **not** set the submaximal share (§4.2).

4.1a Reconstitution: the source, the calibration, and a contradiction the source raises

The 37% / 65% / 86% W' -recovery at 2 / 6 / 15 min (half-time 234 s) is **Ferguson et al. 2010** (*J Appl Physiol*, DOI 10.1152/japplphysiol.91425.2008), reviewed by Chorley & Lamb 2020, who state it was measured "when recovering at a **nominal 20 W**." Three things follow.

- **τ_g is calibrated to ~470 s, not confounded (but not exactly "passive").** The confound of review 3 — the LT1 gate makes the effective constant τ_g/gate , so the fit depends on recovery power — is resolved by the 20 W figure. But 20 W is not quite $\text{gate} = 1$: with $\text{LT1} \approx 0.8 \cdot \text{CP} \approx 200 \text{ W}$, $\text{gate} = (200-20)/200 = 0.90$, so a model reproducing Ferguson needs $\tau_g \approx 470 \text{ s}$ ($\pm$ a few % on the cohort's unreported LT1), **not 520**. The default is corrected to **470**.
- **f_p is consistent with the curve, not independently corroborated by it.** With recovery power fixed, the curve still constrains a **ridge in (f_p, τ_g)** , not f_p alone: $(f_p \ 0.25, \tau_g \ 520)$ and the true joint optimum **$(f_p \ 0.20, \tau_g \ 470)$** (SSE $\approx 10.7 \text{ pts}^2$) both fit 37/65/86 to within ~3–5 pts. So the honest statement is *one physiological estimate of f_p (di Prampero & Ferretti 1999, alactic O_2 -debt $t_{1/2} \approx 30 \text{ s}$; Bangsbo 1990, ~20% alactic) plus a reconstitution curve consistent with it* — the physiology is what selects $f_p \approx 0.25$ on the ridge. We keep $f_p = 0.25$ as the physiological midpoint but note the recovery data alone would prefer ~0.20. (*An earlier draft called this "two independent lines converge"; that overstated the independence — corrected here.*)
- **Ferguson's own components look ~4× off from the model's — but the offset is forced by the data, and once you see why, it exonerates the τ 's and convicts the naming.** This is the most important thing in the paper, and v0.7 reads it correctly (Reviewer 2, third report, was right and v0.6 was wrong). Ferguson reports the two recovery channels underlying W' reconstitution: $\text{VO}_2 \ t_{1/2} = 74 \text{ s}$ (a proxy for oxidative/PCr resynthesis) and blood-lactate $t_{1/2} = 1366 \text{ s}$. Put those beside the model's component half-times:

Component	Model	Ferguson channel	ratio
fast	$\tau_p = 27 \text{ s} \rightarrow t_{1/2} \ 18.7 \text{ s}$	$\text{VO}_2 \ 74 \text{ s}$	4.0×
slow	$\tau_g = 470 \text{ s} \rightarrow t_{1/2} \ 326 \text{ s}$	lactate 1366 s	4.2×
ratio (slow/fast)	17.4×	18.5×	within 6%

Two facts. **(1) The shape is right; only the scale is off, by the same factor in both channels.** The model reproduces the 18-fold fast/slow separation almost exactly — that is not what fitting-two- exponentials-to-a-sum looks like (that lands one component anywhere and lets the other absorb the residual). **(2) The offset is unavoidable, because W' recovers ~4× faster than its own metabolites.** Build the two-compartment model out of Ferguson's *measured* channel constants (107 s, 1971 s) and it cannot reproduce Ferguson's *own* recovery curve:

	2 min	6 min	15 min
Observed (Ferguson 2010)	37%	65%	86%
Model ($\tau_p \ 27$, $\tau_g \ 470$)	40%	63%	87%
Built from Ferguson's channels (107 s, 1971 s)	20%	35%	50%

A model calibrated to the *metabolites* predicts half the recovery that is actually observed. So the τ 's are **not** a fitting error — they are the constants that reproduce W' recovery, which is the quantity the model tracks and

displays, and W' recovery is simply $\sim 4\times$ faster than the PCr/lactate it correlates with. This is Ferguson's own conclusion — W' reconstitution is "**not a unique function of phosphocreatine concentration or arterial [lactate]... unlikely to simply reflect a finite energy store**" — now with a number attached. **What fails is the metabolic naming, not the time constants.** The tanks are compartments of W' (right internal ratio, $\sim 4\times$ -fast absolute scale); the "PCr"/"glycolytic" labels are motivation. Note the symmetry v0.6 missed: *blood* lactate $t_{1/2}$ is no more a measure of *muscle* H^+/Pi clearance than VO_2 off-kinetics is a measure of PCr resynthesis — so **neither Ferguson channel measures a tank, and Ferguson therefore cannot adjudicate the decomposition either way**, which is exactly Ferguson's point and ours. The rename in §4.1 (fast/slow reserve) is the resolution.

- **§6.9's training-load claim does not inherit this problem, and here is why (Reviewer 2's collision, and its resolution).** A careful reader will worry that §4.1a undercuts §6.9: if the "PCr tank" is not PCr, does "76% fast-reserve" still license "cheap to recover from" (§1)? The answer is that §4.1a is entirely about the **recovery law** (τ_p , τ_g), while §6.9's signal is entirely about the **depletion law** — which is anchored in *measured* metabolic kinetics: τ_{on} from Parolin's phosphorylase activation (a direct biopsy measurement) and g_{rate} from di Prampero's flux ratio. Those are not fitted to W' . Concretely, re-running §6.9 with τ_p set to **107 s** (Ferguson's own VO_2 channel — the value that supposedly indicts the model) moves the alactic/glycolytic separation by <1 pt ($77.1\% \rightarrow 75\%$). The separation never collapses across any assumption it rests on (§6.9). So the depletion-side partition can be metabolically meaningful even though the recovery-law naming is not — the training-load argument is best stated in terms of *measured activation kinetics*, not tanks.

4.2 The 1 Hz update

Let P be instantaneous power (W) and $\Delta t = 1$ s. Define the demand relative to aerobic supply as $\Delta = P - CP$.

Case 1 — depletion ($\Delta > 0$, above CP). Demand $need = \Delta \cdot \Delta t$ (J) is met by **both systems in parallel**.

This matters: glycolytic ATP supply is not instantaneous. Glycogen phosphorylase (the rate-limiting glycolytic enzyme) transforms to its active form over the first several seconds of maximal effort — Parolin *et al.* (1999) measured it rising from 12% at rest to 47% by 6 s — while phosphocreatine is the immediate buffer (a fresh 30 s sprint drains muscle PCr to $\sim 17\%$, Bogdanis *et al.* 1996; fresh 10 s and 20 s maximal sprints leave substantially more, Bogdanis *et al.* **1998**); both contribute from the onset, not in sequence (González-Alonso *et al.* 2000). (*The model's own ~ 20 – 30% residual after a 10 s sprint is a separate quantity — a fast-reserve-of- W' level, §4.4 — not muscle PCr; v0.6 mis-cited Bogdanis 1996's fatigued sprint-2 datum for a fresh 10 s sprint. Corrected, and Bogdanis 1998, which reports fresh 10/20 s conditions, is added.*) We model this with a **glycolytic activation** term $g \in [0, 1]$ that ramps first-order with a time constant $\tau_{on} \approx 6$ s while above CP and relaxes back during recovery:

```

g ← g + (1 - g) · (1 - exp(-Δt/τon))      # glycolytic activation ("inertia")

# --- SHARE of submaximal demand: capacity-proportional (NOT rate-proportional) ---
wp = Cp                                # at g=0 → PCr covers all; at g=1 → PCr cov
wg = Cg · g
sharep = need · wp / (wp + wg)      # (guard: if wp+wg ≈ 0, sharep = need)
shareg = need - sharep

# --- RATE CEILING: peak flux, PCr tapered with fullness. Governs MAXIMAL efforts only
ratep = Ppmax · (Rp / Cp)          # (guard: Cp > 0)
rateg = grate · Ppmax · g

takep = min(sharep, Rp, ratep·Δt)      # each tank capacity- AND rate-limited
takeg = min(shareg, Rg, rateg·Δt)
unmet = need - takep - takeg
# shortfall (a tank empty or rate-capped) spills to the partner, then to a deficit:
takeg += min(unmet, Rg - takeg, rateg·Δt - takeg); unmet -= ...
takep += min(unmet, Rp - takep, ratep·Δt - takep); unmet -= ...
Rp -= takep; Rg -= takeg
D += unmet                                # DEFICIT (debt): supra-CP work the caps co

if unmet > 0: rate_limited = true          # producing power beyond the tanks' flux -

```

Why the share and the ceiling are different objects. The rate ceiling — P_{p_max} , g_{pmax} — encodes "PCr is the higher-power system." That is true at **maximal** effort, where both systems are flux-limited, and it is where PCr dominance belongs. It is the **wrong** object for apportioning **submaximal** supra-CP demand, where neither system is near its flux limit and the sharing is set by metabolic control, not peak-flux ratios. Earlier revisions used one rate-weighted rule for both jobs, which forced the PCr trajectory to be a fixed, convex, intensity-*invariant* function of W' -spent (PCr at ~9% by the midpoint of *any* hard effort). Weighting the **share by capacity** and keeping the taper on the **ceiling** fixes it with no cost in the sprint case (there the caps and spill dominate and the share rule never binds — verified bit-identical):

- **Submaximal:** both tanks drain $\approx$ proportionally to capacity, so $R_p/C_p \approx R_g/C_g \approx W'_{bal}$ and both reach their nadir *together* at exhaustion — matching the reproducible-metabolic-milieu picture in §2. Honestly, this means during a steady supra-CP effort the two bars track W'_{bal} and each other; they diverge only in the activation ramp and in recovery. That is not a defect — it is the model behaving the way the Scope box and §7 already say it does (extra resolution lives in transients).
- **Maximal:** the tapered ceiling makes R_p decay ~geometrically to its nadir at exhaustion (a ~10 s all-out sprint leaves the fast reserve at ≈ 20 –30%; §4.4), and fast-reserve dominance emerges from the ceiling, not the weight. (*We no longer claim this "matches Bogdanis 17%": since §4.1a establishes the*

fast reserve is not muscle PCr, we cannot invoke the muscle-PCr mapping to validate the residual and then disown it elsewhere — the residual is a model output, motivated by, not validated against, the biopsy value. Reviewer 3's "you can't have the mapping both ways" is correct and applied here.)

- **Energy conserved even when a cap binds.** Residual `unmet` is banked in a **deficit** `D` (standard `W'bal` permits a negative balance), so $(R_p + R_g - D)$ drops by exactly $\Delta \cdot \Delta t$ per second and §6.2 holds. `D` clears only below LT1 (Case 2) — it is supra-cap byproduct load and must respect the same intensity gate as the glycolytic tank. Note the deficit preserves the *aggregate* but not the *split*: work booked to `D` never drains a tank, so if `D` is large the two bars read slightly full. `D` is an accounting term for the rare supra-flux case, usually signalling a **stale** `P1s`, not fatigue.

On identifiability (and consistency with §7). The parallel draw is the physiologically faithful choice for the *live display*, and `τ_on` is a literature-set constant (Parolin 1999), not a fitted one. But we are careful not to over-claim: the depletion split is **not** identifiable from power at all — total draw above CP is all the pedals see, and any (f_p, g_{pmax}) that sums to the same total is observationally equivalent. In *principle* the bi-exponential **recovery** of `W'` constrains the split (a fast PCr component + a slow glycolytic component); in *practice* that fit is ill-conditioned (§6.3 shows the fast component is invisible at standard reconstitution sampling), so `f_p` ends up **assumed and weakly constrained**, not recovered. This is the §7 position, and it is the one to trust — the reserves are a plausible decomposition, not measured latent state.

Case 2 — restoration ($\Delta \leq 0$, at/below CP). Recovery "headroom" is `CP - P`. Each tank refills exponentially toward full; glycolytic refill is gated by intensity.

```
g ← g · exp(-Δt/τ_off) # glycolytic DE-activation (τ_off =

# FAST RESERVE: resynthesis needs aerobic ATP ABOVE the ride's own demand, so it is ga
# OXIDATIVE HEADROOM (CP - P): full at rest, near-arrested at CP. gate_p scales the AM
# (not the exponent) — the pseudocode and both codebases agree on this (closes Reviewe
gate_p = max(0, (CP - P) / CP)
R_p += gate_p · (C_p - R_p) · (1 - exp(-Δt/τ_p))

# SLOW RESERVE and the deficit recover whenever P < CP, at Skiba's intensity-dependent
# τ_W'(D_CP) = 546·e^(-0.01·(CP-P)) + 316, its amplitude re-anchored so the 20 W passi
# Ferguson 2010 (as v0.6's linear gate did at that one point). NEW in v0.7 — replaces
# (LT1-P)/LT1 gate, whose rate went to zero at LT1 (recovery → ∞), which made the mode
# complete a standard 4×4 (§4.4). Skiba's form is bounded and was fitted ACROSS recove
if P < CP:
    f = gate20 · τ_W'(CP - 20) / τ_W'(CP - P) # =gate20 at 20 W (Ferguson anchor),
    b = f · (1 - exp(-Δt/τ_g))
    R_g += (C_g - R_g) · b
    D -= D · b
```

The v0.7 recovery fix — the linear LT1 gate is gone. v0.6's glycolytic/deficit recovery used a linear gate $(LT1 - P)/LT1$ that fell to zero at LT1, so the effective recovery constant diverged to infinity as the recovery power approached LT1. Re-implementing the pseudocode and holding a rider at constant recovery powers (Reviewer 3, M2) showed the cost: W'bal half-times of 675 s at 150 W and 2601 s at 190 W (vs Skiba's 351 s and 417 s), and — decisively — **the model could not get a rider through $4 \times [4 \text{ min @ } 330 \text{ W} / 4 \text{ min @ } 150 \text{ W}]$** , driving W'bal to 100/38/-23/-76% and leaving the rider 76% into deficit from rep 3 onward. Riders complete that session every week. The fix is Reviewer 3's own suggestion: use Skiba's $\tau_{W'}(D_{CP})$ — a functional prior *already fitted across recovery powers* — for the shape, and re-anchor its amplitude so the 20 W passive rate still reproduces Ferguson (the one curve the model is calibrated to). Result (verified): W'bal half-times are now **201 / 241 / 291 / 367 s** at 20 / 100 / 150 / 190 W — bounded, monotone, and within ~15% of Skiba everywhere — the Ferguson curve is unchanged (40/63/87), the §4.4 sprint battery is bit-identical, and the **4×4 now completes: 100/49/7/-24%** (dips negative only going into the final rep, consistent with a genuinely hard but finishable $\dot{V}O_{2\max}$ session).

The oxidative-headroom gate $gate_p$ (v0.5) — a good fix, and now flagged as the assumption it is. $gate_p = (CP - P)/CP$ stops the fast-reserve bar refilling while the rider is under load (before it, the bar went green holding 99% of CP). It is correct in instinct and fixes a real bug. But (a) its **shape is a choice**: linear-in-CP is one option; a $\sqrt{}$ or sigmoid form is equally defensible and swings the headline 60-s recovery number by **21–31 points** at LT1 (linear 46% / $\sqrt{}$ 65% / sigmoid 34% after a 700 W/20 s effort) — so it is now in the §7 assumptions list with that sensitivity reported. And (b) it only fixes *half* the scenario it was built for: the gate scales the *refill*, it adds no sub-CP *drain*. A rider who sits at $0.95 \cdot CP$ in a bunch for 30 minutes — never above CP — keeps a **100%** fast-reserve bar the whole time, because $\Delta \leq 0$ and the depletion branch never fires. The structurally correct fix is to relax R_p toward an intensity-dependent steady state $R_p^*(P)$ rather than toward C_p ; until then this is a stated blind spot, not a solved case (Reviewer 3, M7b). Verified after a 20 s / 700 W effort (fast reserve $\rightarrow$ 23%), 60 s recovery restores it to **87% at 0 W, 46% at $0.8 \cdot CP$, 25% at $0.99 \cdot CP$** — vs a flat 92% before the gate.

The g de-activation is what makes the activation ramp re-fire on each interval of a repeated-bout set (without it, g would ratchet to 1 on the first surge and every later bout would start at a flat split — the ramp inert for the rest of the ride). **$\tau_{off} = \tau_{on}$ is an assumption, and a load-bearing one**: Parolin (1999) measured phosphorylase *activation*, not *de-activation*, which is a separate quantity. If de-activation runs on a minutes timescale rather than seconds, the ramp would not re-fire on 30 s recoveries and the repeated-sprint behaviour changes materially — so τ_{off} is flagged alongside f_p , τ_g (§7).

LT1 still matters, but its role narrowed in v0.7: it sets the **anchor band** of the Skiba recovery shape (via $gate_{20}$), not a hard on/off wall. Recovery now proceeds for any $P < CP$, fastest at low intensity and slowing smoothly through the tempo band — which is more physiological than the old abrupt cutoff and is what fixes the 4×4. LT1 is the rider's **first lactate threshold in watts**, from a threshold test; LT1 and CP vary independently ($LT1 \approx 65\text{--}85\%$ of CP). Where only CP is known, $LT1 \approx 0.80 \cdot CP$ is a fallback, still the weakest default in the model and worth replacing with a measurement.

Optional realism (off by default; the field ships with all three off, so headline numbers are the plain model): - *pH-slowed / fatiguing PCr recovery* (*fatK*): scale τ_p upward as the slow reserve is emptier, $\tau_{p_eff} = \tau_p \cdot (1 + k \cdot (1 - R_g/C_g))$. Note it **cannot** be pushed to reproduce Bogdanis's muscle-PCr recovery (78.7% at 3.8 min) without breaking the Ferguson *W'* curve and the 4×4 — the $k \approx 16$ that hits Bogdanis drops Ferguson to 30/68/94 and the 4×4 back to -30% — which is §4.1a's point restated: *W'* and muscle PCr do not share a time constant. - *Aerobic ramp* ($\tau_{aer} \approx 25$ s): a first-order aerobic supply $A(t)$ rising toward $\min(P, CP)$, with $need = (P - A) \cdot \Delta t$, reproducing the onset O_2 deficit. It is also what absorbs the shortfall across repeated sprints (§6.10). - *Glycolytic flux fatigue* (*g_fat*, **new in v0.7**): scale the glycolytic ceiling down as the slow reserve empties, $rate_{g_eff} = rate_g \cdot (R_g/C_g)^{g_fat}$, capturing acidotic inhibition of phosphorylase/PFK. Off (0) by default. It moves the repeated-sprint partition from *rising* toward *flat/falling* — the biopsy direction — but does not reproduce the 40% → <10% magnitude (§6.10).

4.3 Reported quantities

- **PCr reserve** = R_p / C_p (0–100%) — how much "punch" is left.
- **Glycolytic reserve** = R_g / C_g (0–100%) — how much "sustained dig" is left.
- **Consumption rate** (per system, *W*) — $take_p/\Delta t$, $take_g/\Delta t$, the live draw on each tank.
- **Combined *W'*bal** = $(R_p + R_g - D)/W'$ — the deficit term keeps this **energy-conserving in depletion** (it drops by exactly $\Delta \cdot \Delta t$ per second) so it matches a single-tank *W'*bal reference on above-CP segments. It intentionally **diverges in recovery** (bi-exponential + LT1 gate; §4.4/§6.2), so it is *depletion-compatible*, not identical everywhere.
- **Two distinct flags** (the split the code has always made, now stated in the paper — closes the round-4/round-5 open item):
- **rate_limited** — the rider is producing power beyond the tanks' *flux* caps ($unmet > 0$, *D* growing) while the reserves are **not** empty. Usually signals a stale P_{1s} , not a spent rider; the field surfaces a "recalibrate sprint power" hint.
- **exhausted** — both reserves are at/near zero ($R_p + R_g \approx 0$). This drives the red state. These are different conditions (a fresh rider with a mis-set P_{1s} can be **rate_limited** but not **exhausted**); v0.6's §4.3 wrongly unioned them under one label.

4.4 Why this behaves correctly

- A short, very hard surge draws mostly from R_p (glycolysis has not yet ramped in), and R_p refills after — *at a rate set by recovery intensity* (§4.2 gate). A maximal ~10 s sprint leaves **$R_p \approx 20\text{--}30\%$** , **not empty**, from the tapered ceiling's emergent depletion constant $\tau_{dep} = C_p/P_{p_max}$. **But τ_{dep} is not a fixed 7 s** — §7 notes it swings ~4–13 s across riders, so the 10 s residual ranges from ~10% (fast riders) to ~46% ($\tau_{dep} \approx 13$ s). This is a fast-reserve-of-*W'* level, *motivated* by the biopsy picture (Bogdanis 1996: muscle PCr $\approx 17\%$ at 30 s) but not *validated against* it (§4.1a: the fast reserve is not muscle PCr); the wide across-rider spread is an argument for a *measured* depletion constant (the re-architect fork, §7), not an emergent ratio.

- A sustained supra-CP effort draws from **both** tanks, declining together to their nadir at exhaustion. The drawdown is **front-loaded (mildly concave), not linear** — the activation ramp makes the early draw PCr-weighted (450 W: 68/42/18/1 at quarter-TTE intervals). During such steady efforts **both bars are affine in W'bal** (§4.2), so they carry no information single-tank W'bal lacks *there*; their only non-redundant content is the ramp offset (set by $\tau_{\text{on}} / \tau_{\text{off}}$) and the post-effort PCr recovery transient. This is stated as a limitation, not a feature.
- **A falsification test (not a validation), and its real cost.** For a constant supra-CP effort, R_p/C_p is *asymptotically* a function of the fraction ϕ of W' spent — but only for efforts long relative to τ_{on} and below the ceiling regime; the activation ramp adds a genuine second intensity dependence (a hard effort reaches a given ϕ faster, with g lower, so PCr is *lower* at that ϕ). So the test "hold different supra-CP powers to exhaustion and compare PCr-at-%W'-spent" would report intensity-*dependence* for a correct model, and a naive read would wrongly "falsify" it. Framed correctly it is a **falsification** test (pass $\Rightarrow$ the bar is a deterministic restatement of W'bal in depletion, i.e. redundant; fail $\Rightarrow$ the architecture is wrong) — lose-lose for the *depletion* claim, which is the point. And it is **not** "testable in an afternoon": checking PCr in-cycling needs 31P-MRS, which §6.5 says is unavailable in this modality. It requires a surrogate (post-exercise sampling, or knee-extension 31P-MRS with a cross-modality transfer assumption; §8).
- **Synthetic battery (v0.5).** Parameters: $CP = 255 \text{ W}$, $W' = 20 \text{ kJ}$, $f_p = 0.25$, $P_{p_max} = 690 \text{ W}$ ($g_{\text{rate}} = 0.5$), $\tau_p = 27$, $\tau_g = 470$, $\tau_{\text{on}} = \tau_{\text{off}} = 6$, $LT1 = 204 \text{ W}$.

Test	Result
Sustained 450 W — PCr at 25/50/75/100% of TTE	68 / 42 / 18 / ~1% (front-loaded, nadir at exhaustion)
Both tanks at exhaustion	PCr ~1% / GLY ~0% (empty together)
945 W (= P_{1s}) 10 s sprint — PCr residual	23% (ceiling-dominated; share rule doesn't bind)
6×[10 s@700 W / 30 s@150 W] — PCr at each bout end	49 → 17 → 10 → 8 → 8 → 7% (gated recovery: 150 W is high sub-CP, so little refill)
1200 W hold (caps bind) — combined W'bal conservation	leak = 0 J
Fast-reserve recovery after 700 W/20 s, 60 s @ {0, 128, 204, 242, 252} W	87 / 68 / 46 / 30 / 25% (headroom gate)
Debt repayment at 0.9·CP	D clears slowly (Skiba rate at that power), not frozen

- **Recovery-power behaviour (v0.7 fix, verified).** W'bal half-time after full depletion, holding a constant recovery power — the linear LT1 gate (v0.6) vs the Skiba-shaped gate (v0.7) vs Skiba's own $\tau_{W'}$:

Recovery power	v0.6 linear gate	v0.7 Skiba-shaped	Skiba $\tau_{W'}$
20 W (Ferguson anchor)	201 s	201 s	255 s

Recovery power	v0.6 linear gate	v0.7 Skiba-shaped	Skiba $\tau_{W'}$
100 W (39% CP)	353 s	241 s	299 s
150 W (59% CP)	675 s	291 s	351 s
190 W (75% CP)	2601 s	367 s	417 s
204 W (LT1)	∞	409 s	446 s

The v0.6 column diverges 2–7× from Skiba above 100 W and hits ∞ at LT1; v0.7 tracks Skiba within ~15% everywhere and is bounded. Consequently the **4 × [4 min @ 330 W / 4 min @ 150 W]** session, which v0.6 could not complete (W'_{bal} 100/38/–23/–76%), now reads **100/49/7/–24%** — finishable, as it is in life.

- **On "generalizing W'_{bal} " — a careful claim.** In *depletion* the model is a faithful generalization: $(R_p + R_g - D)$ falls by exactly $\Delta \cdot \Delta t$ per second (cap-binding or not), so the summed draw above CP matches standard W'_{bal} . In *recovery* it is deliberately **different** (bi-exponential, LT1-gated), so it does not reduce to Skiba's single $\tau_{W'}$. It is a generalization in **structure and depletion behaviour**, not a strict superset. (We drop the earlier $f_p \rightarrow 0$ argument: with the fullness taper $C_p = f_p \cdot W' = 0$ is a division by zero, guarded in code — the limit is not meaningfully computable and does not add anything.)

5. Real-time on-device implementation (Garmin Connect IQ)

The model's per-second cost is a handful of multiplies and one `exp()` per tank — trivial for a head unit. A companion document, `connectiq-app-spec-and-prompt.md`, gives the full technical specification and a ready-to-use build prompt. Summary of the target design:

- **App type:** a custom full-screen **Data Field** (`Toybox.WatchUi.DataField`), which the head unit calls once per second — a natural fit for the $\Delta t = 1 \text{ s}$ update.
- **Input:** `Activity.Info.currentPower` inside `compute(info)`.
- **Configuration:** `CP`, `W'`, `f_p`, `P_p_max` (a per-athlete calibration target — §6.7), `LT1` (a measured threshold, not a %CP default — §4.2), and the recovery constants exposed via Connect IQ app settings (`Application.Properties`), so a rider enters them from Garmin Connect.
- **Display:** two vertical bar gauges (PCr and glycolytic reserve) with numeric % and color bands (green → amber → red). *Per-system live consumption in W is a "modelled share," not a reading* — when the ceiling is slack it is just $(\text{power} - \text{CP})$ split by a fixed fraction (§7), so it is labelled as such or omitted.
- **Recording:** write both reserves and consumption to the FIT file via `FitContributor` fields so they sync to Garmin Connect / intervals.icu / Strava for post-ride analysis.
- **Footprint:** a handful of scalar state variables (`R_p`, `R_g`, the activation `g`, the deficit `D`), no arrays — well within Connect IQ memory budgets. Per-second cost is still a few multiplies and one `exp()`

per tank.

6. Validation strategy

The model is a hypothesis; it must be checked against data before its numbers are trusted. The honest difficulty is that the model's *headline feature* — a separate PCr reserve tracked in real cycling — is the hardest thing to validate, so we separate tests that only check the plumbing from the one test that would actually earn the second tank.

1. **Face validity / unit tests** — synthetic power traces (single sprint, repeated sprints, supra-CP hold, sub-CP recovery) should produce the qualitative behaviours in §4.4.
2. **Backward compatibility — depletion only.** The combined balance $(R_p + R_g - D)$ reproduces a reference W'bal implementation (Froncioni–Clarke) on the same trace **in depletion** — it drops by exactly $\Delta \cdot \Delta t$ per second, and the deficit term D preserves that even when a rate cap binds. It will **not** match in recovery, by construction (§4.4), so this test must be scoped to above-CP segments — a test that expected agreement everywhere is one the model is designed to fail.
3. **Reconstitution targets — what the curve constrains, and what it cannot.** Combined recovery after a full W' depletion is compared to the published curve (37 / 65 / 86% at 2 / 6 / 15 min; Ferguson et al. 2010, at 20 W recovery). (a) **The curve is blind to the fast tank** — a $\tau_p \approx 27$ s process is $1 - e^{(-120/27)} \approx 99\%$ complete by the first (2 min) sample, so it validates *glycolytic* recovery and the size of the PCr offset, **not** PCr kinetics (for those you need 10/20/30/45/60 s samples; test 4). (b) **The curve fixes τ_g and constrains a (f_p, τ_g) ridge, not f_p alone.** With the 20 W recovery power known (gate ≈ 0.90), the fit gives $\tau_g \approx 470$ and the joint optimum $(f_p \approx 0.20, \tau_g \approx 470)$; f_p is selected on that ridge by physiology, not by the curve (§4.1a). So the curve is **consistent with $f_p \approx 0.20\text{--}0.25$** , not an independent measurement of it. (This supersedes a v0.3 statement — since deleted — that computed an implied $f_p \approx 0.13$ at the old $\tau_g = 360$ s; that value was a fossil of the superseded default.)
4. **Fast-recovery kinetics (the missing piece).** The load-bearing test for τ_p is a repeated-bout protocol with **early** recovery sampling — e.g. all-out efforts separated by 10/20/30/45/60 s — and the between-bout power recovery correlated with modelled PCr recovery (as Bogdanis 1996 did for PCr). Standard reconstitution data cannot supply this; it must be measured deliberately.
5. **System-specific truth (lab) — and its hard limit.** Blood-lactate kinetics can corroborate the glycolytic tank, but the PCr tank's gold standard, **31P-MRS, is essentially unavailable in real cycling**: the measurement requires the muscle to be inside a magnet, so it exists only for small-muscle knee-extension or immediate post-exercise, not whole-body cycling. The model's headline novelty therefore **cannot be checked against the gold standard in the modality it targets** — a limitation to state plainly, not paper over.
6. **The test that earns the second tank — and what it actually tests.** Because the model is a near-superset of single-tank in depletion, "it fits better" is guaranteed and is **not** evidence the compartments are real. The decisive test is a **head-to-head against single-tank W'bal on an out-of-sample**

intermittent-tolerance prediction (the protocol class where the hydraulic model beat W'bal, arXiv 2108.04510). But be precise about what it probes: since this model's *depletion* is single-tank by construction, its **only** lever to beat single-tank is the **recovery law** (bi-exponential + LT1 gate). So **test 6 is a test of the recovery law, not of the two-compartment hypothesis** — the framing "two tanks → better prediction" overstates what a win would show. This also sharpens the protocol: choose interval recovery-valley powers that **straddle LT1**, where the dual-tank and single-tank recovery laws diverge most, rather than a generic interval set.

7. **Field calibration — scoped honestly.** Fit **P_pmax** and the **τ**'s per athlete to maximal and repeated-bout tests. **f_p is not a routine calibration target:** §4.2 shows the depletion split is unidentifiable from power, and only the test-4 class (all-out efforts with early recovery sampling) can constrain it — and even then the fit is ill-conditioned. Also: any single-tank W'bal comparator in test 6 must have **its own τ_{W'} fit to the same data**, or the dual-tank wins trivially by having more recovery freedom and the win means nothing.

6.8 First empirical results (v0.5) — usefulness, on real rides

Three reviews converged on the point that the paper validated *plausibility* but never the two cheap, data-in-hand questions that decide whether the feature is worth building. Both were run on six real interval/VO₂max sessions (the model's best case — a steady endurance ride would score far lower):

- **How often is the fast-reserve bar informative, and is it a function of W'bal?** Two tests. (a) The bar is below 95% (i.e. live) for a per-ride **20 / 31 / 38 / 52 / 61 / 79%** of ride time — median ~45%, range 20–79% (reported per-ride, not as a bare mean, per Reviewer 2 §8). (b) The stronger test, which a reviewer set and the bar passed: is the bar just an affine transform of W'bal? Binning W'bal and measuring the spread of **R_p/C_p** within each 1-point bin gives a **mean spread of ~58 pts (max ~93)**, and regressing the bar on its best affine W'bal predictor gives **R² ≈ 0.03** — the bar sits >5 pts off that affine line ~93% of the time. So the fast-reserve bar is emphatically **not** a function of W'bal; the ~45% figure is, if anything, conservative.
- **Does the depletion machinery change the display, vs the recovery-only null?** Running the full model and the null (§7) on the same traces, the fast-reserve bar **diverges** by up to ~25–60 pts, maxima at the starts of recovery valleys. **Important caveat, now carried up from §7: divergence is not accuracy.** Two models differing by 25–60 pts tells us the caps *change* the display, not that the changed display is *right* — a random process would diverge too. Whether the extra resolution is *correct* is §6.6 and now §6.10 (which suggests some of it is not). §6.8 establishes the bar carries *information*, full stop.

Neither result validates the *compartments* (that is §6.6 and §6.10). They answer a narrower, real question — *is the two-bar display worth showing, and does the model's complexity change what it shows* — with data instead of assertion; the answers are yes and yes, for interval training, with correctness still open.

6.9 Training-load partitioning — a use case the reviews evaluated past

Every review assessed the model as a **pacing** aid and correctly found the live bars affine with W'_{bal} outside transients. But there is a second use case they did not weigh, and it is where the decomposition is most clearly worth more than W'_{bal} : **training prescription and load monitoring**. The relevant output here is not the live bar but the **cumulative per-system load over a session** — how many kJ came from the alactic (PCr) vs the glycolytic system — which the field already records ($PCr_depleted_kJ$, $GLY_depleted_kJ$). This matters because the two systems carry asymmetric recovery costs (§1): alactic work is cheap and repeatable; glycolytic work is expensive and self-limiting. A coach building a PCr-targeted block (alactic power, repeat-sprint ability) wants sessions that stay alactic; one building lactate tolerance wants the opposite. **W'_{bal} cannot tell these apart — it is one number.**

The model does discriminate session types, but v0.6 credited the wrong mechanism and overstated the baseline it beats. Both are corrected here. Sessions are now **fully specified** (Reviewer 2 §8): the alactic archetype is $10 \times [6 \text{ s @ } 900 \text{ W} / 300 \text{ s recovery @ } 150 \text{ W}]$; the glycolytic archetype is $5 \times [60 \text{ s @ } 360 \text{ W} / 120 \text{ s recovery @ } 150 \text{ W}]$. Per-system load (default parameters, g_fat off), now reporting the **deficit D** as **a third number** (work booked to neither tank; Reviewer 3 M4):

Session	fast-reserve kJ	slow-reserve kJ	D kJ	fast-reserve % of (fast+slow)
$10 \times [6 \text{ s @ } 900 \text{ W}]$ — alactic	29.2	8.7	1.2	77%
$5 \times [60 \text{ s @ } 360 \text{ W}]$ — glycolytic	13.2	21.5	1.8	38%

The separation is real (~39 pts). But three v0.6 claims about it were wrong:

(a) The mechanism is the glycolytic flux ceiling scaled by the ramp — not the ramp alone. Ablating the model one part at a time on the alactic session:

Configuration	alactic fast-reserve %	separation vs glycolytic
full model	77%	~39 pts
no activation ramp ($g \equiv 1$)	52%	~15 pts
no rate ceilings (share rule only)	47%	~9 pts
neither	25%	−12 pts

Removing the ceilings costs ~30 pts; removing the ramp ~25 pts; neither alone reaches the full 77%. It is an **interaction**: during a 6 s sprint the glycolytic ceiling ($g_rate \cdot P_p_max \cdot g$) caps glycolysis at a few tens of watts while demand is ~645 W, and the spill sends the difference to the fast reserve. So the load-bearing parameter is **g_rate** (via the ceiling), not τ_{on} / τ_{off} . v0.6's "the discriminating content is the activation ramp" and §7's " τ_{off} is the only parameter carrying the depletion novelty" are both **withdrawn** — a 20× change in τ_{off} moves the split 1–2 pts, whereas g_rate 0.3 → 0.7 moves the separation 48 → 33 pts.

(b) The metric detects supra-flux-ceiling power, so it is blind to submaximal alactic work. Hold the structure fixed (10×6 s, full recovery) and sweep the power: the reading climbs 52% → 62% → 66% → 72% → 77% at 450 / 550 / 600 / 700 / 900 W. A textbook alactic session done at 450 W — short efforts, full recoveries, exactly what a neuromuscular block prescribes — reads ~52%, close to the glycolytic session's 38%. The metric is really answering "did you exceed the glycolytic flux ceiling," not "was this alactic." A coach needs to know this; it is a genuine scope limit, not a footnote.

(c) The honest baseline is not W'bal — it is a stopwatch. "W'bal cannot tell these apart — it is one number" is false: W'bal is a *time series*, and a coach with the power file and no model can compute the fraction of supra-CP work done in efforts under 15 s — **100% vs 0%** for these two sessions, perfect separation, zero parameters (Reviewer 3 M3). So §6.9's real question is not "does the model separate these" (of course it does) but "**does the per-system load carry information beyond effort duration and intensity?**" That is an open, testable question (regress fast-reserve % on a duration/intensity baseline across real rides; if $R^2 > 0.9$ the metric is an expensive stopwatch), and the paper should not claim the win until it is answered.

What *does* survive: (i) the separation is robust to every assumption it rests on — including setting τ_p to Ferguson's 107 s (§4.1a), which moves it <1 pt, so §4.1a does not reach it; (ii) it is a genuinely different functional of the power trace than W'bal. Honest limits: absolute kJ scale with f_p ; D must be reported (and the split refused when D is a large fraction of total — e.g. a 4×4 books ~45% of its anaerobic work to D , so a per-system split there is meaningless); and this is a training-*design* signal, not a validated *outcome*.

6.10 Repeated-sprint ATP partitioning — an in-modality, compartment-level test the model fails

v0.6 declared (§7) that the compartments are "unfalsifiable in-modality — permanently." That is **withdrawn**. The claim rested on two true premises (the split is unidentifiable from power; 31P-MRS cannot run during cycling) and one false inference — that 31P-MRS is the *only* compartment-level measurement. **Muscle biopsy during cycle ergometry is another**, and Gaitanos 1993, Parolin 1999, and *Sci Rep* 2024 — all already cited — partition ATP supply by system from vastus-lateralis biopsies during maximal cycle-ergometer sprints. That is compartment-level data, in the target modality, on the target ergometer, for the exact protocol class the model describes. It needs no new data collection.

So we ran it. The model's own §6.9 alactic protocol (10×6 s maximal sprints) is exactly Gaitanos's, and the biopsy literature reports the glycolytic ATP share **falling from ~40% in the first sprint to <10% in the last**, with aerobic metabolism rising to cover the shortfall. The model, as shipped:

	Sprint 1	→	Sprint 10
Observed (Gaitanos 1993 / <i>Sci Rep</i> 2024), glycolytic share of anaerobic ATP	~40%	↘	<10%

	Sprint 1	→	Sprint 10
Model, shipped (<code>g_fat</code> = 0)	23%	↗	46%
Model, optional <code>g_fat</code> = 1 + aerobic ramp	23%	→	~34% (flat, then falling)

Wrong level and wrong direction. The model starts glycolysis at about half the observed share and has it *rise* across bouts, because it has the two fatigue mechanisms backwards: the fast-reserve ceiling (`rate_p` = `P_p_max` · `R_p` / `C_p`) fades as the reserve empties, so more demand spills to glycolysis bout over bout, while glycolytic flux has *no* fatigue term at all. The optional `g_fat` term (§4.2) plus the aerobic ramp flatten the trend and pull the late bouts down, but they do **not** reproduce 40% → <10% — the starting level is set by `g_rate` and the activation ramp, and reaching 40% in sprint 1 would require a higher `g_rate` that then wrecks the alactic/glycolytic separation of §6.9.

Two conclusions, and both are progress. **First, this is the first genuinely falsifiable result in the project** — every prior round argued internal consistency, identifiability, and framing; this one has a number that is wrong in a specific, fixable way, and it arrived because §6.9 finally asked a question with a checkable answer.

Second, it settles what the tanks are: as shipped, the depletion split is a **W'-decomposition heuristic, not a validated bioenergetic ATP partition**. The `PCr_depleted_kJ` / `GLY_depleted_kJ` fields should be read as a *descriptive statistic* of a power file (how front-loaded and supra-ceiling the anaerobic work was), not as a biopsy-grade estimate of which metabolic system paid. Whether the model can be made to reproduce 40% → <10% — by calibrating `g_rate` to the sprint-1 biopsy value and fitting `g_fat` to the decay — is the single highest-value experiment left, it needs only data already in the bibliography, and it is the one test in this document that could falsify (or earn) the two-compartment hypothesis directly rather than by proxy. Until it is run, §6.9 stands as a descriptive metric, not a bioenergetic one.

7. Assumptions and limitations

- **The PCr/glycolytic split `f_p` is assumed (physiology), and consistent with — not corroborated by — the recovery data.** Power cannot identify the depletion split (§4.2). We default to **0.25** from physiological alactic-fraction estimates (~0.20–0.30; di Prampero & Ferretti 1999, Bangsbo 1990 ~20%). The reconstitution curve constrains a (`f_p`, `τ_g`) ridge, not `f_p`, and physiology is what selects `f_p` on it (§4.1a) — so the two "lines" are not independent, and the recovery data alone would prefer ~0.20. Supported band ~0.20–0.25; a per-athlete target; outputs are sensitive to it.
- **The reserves are a decomposition of W', not latent metabolic state.** The two bars are a physiologically-motivated split of one measured quantity (W'bal), not two independently measured reserves, and §4.1a shows they are compartments of W' rather than of muscle metabolites. **Correcting a v0.6 under-claim:** the fast-reserve bar is *not* "full most of the time" while riding. Under the oxidative-headroom gate the effective recovery constant is `τ_p/gate_p` — **27 s only at a standstill**,

rising to ~44 s at 100 W, ~66 s at 150 W, ~135 s at LT1, and 100s–500s across the sub-CP band a rider actually recovers in. So the bar stays live through interval recovery valleys (§6.8: 20–79% of ride time), and the "affine transform of W'_{bal} " caveat applies to *stopped or easy* recovery, not to riding — which is more favourable to the display than v0.6 stated. When the fast reserve is full (rest), the glycolytic bar is an affine transform of W'_{bal} ($W'_{bal} = f_p + (1-f_p) \cdot R_g/C_g$); the honest product framing is " W'_{bal} plus one new number," a heuristic decomposition whose split is assumed.

- **gate_p (the PCr oxidative-headroom gate) is an assumption with a consequential undefended shape.** It fixes a real bug (§4.2) but its functional form is a choice: linear-in-CP vs $\sqrt{}$ vs sigmoid swing the 60-s post-effort recovery number by **21–31 points at LT1** (46 / 65 / 34%). It sets the headline display number and is flagged here alongside the LT1-band shape. It also adds no sub-CP *drain*, so it does not cover the sustained-sub-CP-in-a-bunch case (§4.2).
- **g_rate = 0.5 is load-bearing, not an aside.** Via the glycolytic flux ceiling it generates §6.9's headline separation (ablation: removing the ceilings costs ~30 pts), and sweeping it 0.3 → 0.7 moves that separation ~48 → 33 pts. It has never been fit to the sprint biopsy data that would constrain it (§6.10). Flagged and banded (~0.3–0.7) alongside f_p .
- **The PCr depletion rate is also assumed — and it was invisible until v0.4.** The tapered ceiling gives an emergent depletion constant $\tau_{dep} = C_p/P_{p_max}$, tied to *no* measured PCr depletion kinetic; it is the ratio of three parameters and swings ~3× across plausible riders (~4–13 s). §7 was scrupulous about the assumed *recovery* constants and silent about this equally load-bearing *depletion* one. It should be sanity-checked against literature PCr depletion half-times, not left to fall out of f_p , W' , and a 1 s sprint power.
- **τ_g is calibrated to ~470 (confound resolved); τ_{off} remains assumed but is not load-bearing.** Ferguson recovered at ~20 W (gate ≈ 0.90), so $\tau_g \approx 470$ (§4.1a). $\tau_{off} = \tau_{on}$ is still an assumption (Parolin measured activation, not de-activation; §4.2). **Retraction:** v0.6 (crediting Reviewer 2's round-2 claim) said τ_{off} "is the *only* parameter carrying the depletion-phase novelty." §6.9's ablation falsifies that — a 20× change in τ_{off} moves the per-system split 1–2 pts; the novelty is carried by the **glycolytic flux ceiling (g_rate) scaled by the ramp**, not by τ_{off} . Reviewer 2 retracted the original claim in the third report; it is removed here.
- **LT1 should be measured, not derived from CP.** The $0.80 \cdot CP$ fallback will be wrong for many riders (LT1 ranges ~65–85% of CP), and it gates whether the glycolytic tank recovers during tempo, so a bad value materially changes recovery estimates.
- **Several small modeling choices are assumptions, flagged not defended.** The glycolytic recovery *shape* (v0.7 uses Skiba's $\tau_{W'}(CP-P)$, amplitude-anchored to Ferguson at 20 W — a defensible prior fitted across recovery powers, but still a choice of functional form); lumping deficit clearance and slow-reserve refill onto the *same* τ_g ; and the spill order (unmet demand spills to glycolytic before PCr). **The spill-order flag is downgraded:** Reviewer 2 tested it and reversing the order changes §6.9 by 0 pts (the glycolytic rate is already exhausted, so there is nothing to spill), so it is inert on the sessions that matter here.
- **The deficit D corrupts the display at maximal effort, and its recurrence mechanism was misstated.** When D is large the two bars read slightly full. **Correction (Reviewer 3 m4):** v0.6 attributed the late-effort recurrence of D to P_{1s} decaying intra-ride — which is backwards (a decaying sprint

means the rider produces *less* against a *fixed* cap). The real mechanism is the **fullness taper**: $\text{rate}_p = P_{p_max} \cdot (R_p/C_p)$ collapses as the fast reserve empties, so the combined flux ceiling drops below what a deep-but-still-sprinting rider produces, and the residual books to D . That is *structural*, not a settings error, so a "recalibrate sprint power" hint (the genuine *rate_limited* -from-full case; §4.3) will not fix it. D should be reported as a third quantity (§6.9) and the per-system split refused when D is a large fraction of total (e.g. ~45% on a 4×4).

- **The product thesis — two bars → better pacing — is untested as a human-factors claim.** §6.8 shows the bar is *informative* often enough to matter on interval sessions, but not that a rider *decides better* with it. The minimum next step is a usability check (even n=1: race on it, log whether the second bar changed a call) before the display's value is asserted, not just its informativeness.
- **$C_p = 0$ means the usable alactic store is spent, not that muscle PCr is zero.** Real PCr bottoms out around 20–40% of resting at exhaustion; C_p is the usable reserve above that floor, so $R_p = 0$ on the display is "usable punch gone," not a muscle state that never occurs.
- **The compartments are *not* permanently unfalsifiable in-modality — retracted (§6.10).** v0.6 argued from two true premises (the split is unidentifiable from power, §4.2; 31P-MRS cannot run during cycling, §6.5) to a false conclusion, by assuming 31P-MRS is the only compartment-level measurement. **Muscle biopsy during cycle ergometry is another, it is already cited, and the model fails against it** (§6.10: repeated-sprint glycolytic share, model 23% ↗ vs observed 40% ↘). So there is a depletion-side, compartment-level, in-modality validation route, it needs no new data, and it is the highest-value test in the document. Cross-modality 31P-MRS transfer remains a route for the *recovery* constants, but it is no longer the only path. On *correctness* vs single-tank W_{bal} there is still no positive evidence (§6.6); §6.10 supplies the first *negative* one, which is progress.
- **Does the depletion-side machinery earn its keep, and where? (Corrected.)** The null to beat is a single reserve depleting as W_{bal} and recovering bi-exponentially (f_p , fast τ_p , slow τ_g) — no caps/spill/deficit. v0.6 concluded the machinery earns its keep "in exactly one regime, the maximal one" and is "inert everywhere the ceiling does not bind." **§6.9's ablation shows the opposite for short efforts:** the glycolytic ceiling binds *constantly* on 6-s sprints (glycolysis capped at tens of watts vs ~645 W demanded) and is the engine of the training-load separation — so the ceiling is not a sprint-only term, it is load-bearing whenever efforts are short. Where the ceiling is slack (submaximal supra-CP), the model does collapse to the null: $\text{take}_p = \Delta \cdot C_p / (C_p + C_g \cdot g)$ contains neither reserve and settles to a fixed $\Delta \cdot f_p$ split within ~20 s — (*power – CP*) *rescaled* — so **per-system live consumption is a "modelled share," not a reading**, and is labelled as such (§5). Net: the depletion machinery earns its keep in the two regimes where efforts are short or maximal (sprint fidelity, training-load partitioning), and is redundant in the steady submaximal regime — a sharper, ablation-backed answer than v0.6's.
- **On identifiability, name the degenerate directions before fitting (Reviewer 3 m5).** f_p and P_{p_max} both set the sprint residual; τ_p and the pH coefficient k both set the repeated-bout decay; g_{rate} and the activation ramp both set the §6.9 partition level. A per-athlete fit that ignores these degeneracies will fit noise and call it calibration (§6.7 restricts routine fitting to CP , W' , P_{p_max} and the τ 's for this reason).

- **A structural fork the paper now states rather than drifts through (Reviewers 3 and round-5).** The entire lightweight-model rationale (§3's parameter economy, §5's one `exp()` per tank, literature-fixed constants) exists to serve the *pacing* use case on a head unit. But §6.9–§6.10 have moved the paper's centre of gravity to the **training-load** use case — and that is a *post-hoc computation on a power file*. It does not need a head unit, 1 Hz, or nine literature-fixed parameters; it needs `g`, the capacity share, and the flux ceiling, and it can be computed off-device in a few dozen lines. Off-device, §3's argument reverses: with a laptop one can afford the 8-parameter hydraulic model with per-athlete fitting (which already out-predicts W'bal on intermittent protocols) or the 17-parameter bioenergetic model, and calibrate `g_rate / g_fat` to the biopsy data of §6.10. So there is a genuine product decision, stated here for the reader (and the maintainer) to make deliberately rather than by attrition: **if the live two-bar pacing display is the product, the pacing case has to carry it, and that case runs on the fast-reserve recovery transient; if the per-session training-load metric is the product, it belongs in a post-ride tool (intervals.icu / GoldenCheetah), where a better-resolved model is available.** The two are not exclusive, but they have different centres of gravity, and v0.6 blurred them.
 - **Fixed time constants** ignore the documented pH-dependence and bout-to-bout slowing of PCr recovery unless the optional fatigue term is enabled.
 - **Hard CP boundary** omits the aerobic slow component and onset kinetics unless the optional aerobic-ramp term is enabled; expect over-attribution to anaerobic tanks in long low-intensity recovery (the same failure mode reported for the EJAP 2023 model).
 - **Power is whole-body and mechanical**, not muscle-specific; the model cannot see local fatigue, cadence, or fiber-type effects.
 - **Not medical or diagnostic.** This is a training/pacing aid; tank levels are estimates, not measurements of muscle chemistry.
 - **Parameters drift** with fitness, fatigue, heat, and altitude; periodic re-testing is required.
-

8. Conclusion

The single-tank W' model earned its place by being power-native and light enough to run anywhere, but it answers "how much anaerobic work is left?" without answering "in which system?" The dual-tank model proposed here offers a physiologically-motivated split at negligible computational cost: divide W' into a fast, rate-limited, fast-recovering PCr tank and a slow, large, slowly-recovering glycolytic tank; share supra-CP demand between them by **capacity**, cap each by its **peak flux** (PCr tapering with fullness), and refill them with system-specific, intensity-gated laws. It generalizes W'bal in depletion and diverges from it — deliberately — in recovery, it runs at 1 Hz on a Garmin head unit, and it surfaces a two-system decomposition — *punch left* and *dig left* — that is most informative in the transients where pacing decisions concentrate.

v0.7 changes two things structurally and settles a third. **Structurally:** the recovery law is fixed — the linear LT1 gate that made the model unable to complete a 4×4 is replaced by Skiba's intensity-dependent shape (§4.2/§4.4), so recovery is now bounded and monotone across powers, at no cost to the Ferguson curve or the

sprint battery. And the tanks are reframed, following the review series' convergent finding, as **compartments of W' rather than of muscle metabolites**: W' recovers $\sim 4\times$ faster than the PCr and lactate it is named after (§4.1a), which *exonerates* the time constants (they are the right constants for W') and *convicts* the metabolic naming (the labels are motivation, not measurement). Every "the bar equals biopsy PCr" reading is withdrawn.

The settled third thing is the honest status of the whole enterprise. There is still no evidence the decomposition is more *correct* than single-tank W'bal for pacing (that needs §6.6). But v0.6's claim that the compartments are permanently unfalsifiable is **wrong**: §6.10 runs the model against biopsy repeated-sprint data — in-modality, already cited — and it **fails in level and direction** (glycolytic share 23% ↗ vs 40% ↘). That is the first falsifiable result in five rounds, and it reclassifies the output: **the per-system split is a W'-decomposition heuristic and a descriptive statistic of a power file, not a validated bioenergetic ATP partition**. The §6.9 training-load use case remains the strongest reason to build the second tank — but its mechanism is the glycolytic flux ceiling scaled by the ramp (not the ramp alone), it is blind to submaximal alactic work, and it must beat a zero-parameter stopwatch, not just W'bal. So the model is now correctly located: a lightweight, honest, *heuristic* decomposition of W', with a live pacing display whose value rests on the fast-reserve recovery transient, and a post-ride training metric that is promising, descriptive, and — for the first time — has a data-in-hand test that can earn or sink it (§6.10). The physiological framing has run its course; the remaining moves are the recovery-law head-to-head (§6.6) and the biopsy calibration (§6.10), both experiments, not arguments.

References

See `docs/literature-review-anaerobic-models.md` for the full annotated bibliography with DOIs and PMIDs. Primary sources for this paper:

- Skiba PL et al. 2012, *Med Sci Sports Exerc* — W'bal integral model. PMID 22382171, DOI 10.1249/MSS.0b013e31824cfdc0.
- Skiba PL et al. 2015 — differential W'bal. DOI 10.1249/MSS.0000000000000226.
- Bartram J et al. 2018, *IJSPP* — elite $\tau_{W'}$ recalibration. DOI 10.1123/ijsp.2017-0356.
- Chorley A, Lamb K 2020, *Sports (Basel)* — CP/W' reconstitution review (states the 37/65/86% figures were at "nominal 20 W" recovery). DOI 10.3390/sports8090123 (PMC7552657).
- Ferguson C, Rossiter HB, Whipp BJ, Cathcart AJ, Murgatroyd SR, Ward SA 2010, *J Appl Physiol* 108(4):866–874 — W' reconstitution 37/65/86% at 2/6/15 min (half-time 234 s) at 20 W recovery; component recovery channels VO_2 $t_{1/2}$ 74 s and lactate $t_{1/2}$ 1366 s; W' recovery is *not* a unique function of PCr or lactate. PMID 20093659, DOI 10.1152/jap.01425.2008 (the .2008 DOI stem is the submission year; the article is 2010).
- Morton RH 1986, *J Math Biol* — three-component hydraulic model. DOI 10.1007/BF01236892.
- Weigend F, Behncke, Skiba 2021 — hydraulic model & `three_comp_hyd`. arXiv 2104.07903 / 2108.04510.

- Dynamic bioenergetic model, intermittent cycling, *Eur J Appl Physiol* 2023. PMID 37369795, DOI 10.1007/s00421-023-05256-7 (PMC10638188).
- di Prampero PE, Ferretti G 1999, *Respir Physiol* — anaerobic energetics reappraisal. PMID 10647856, DOI 10.1016/S0034-5687(99)00083-3.
- Harris RC et al. 1976 — biphasic PCr resynthesis. *Pflügers Arch*.
- Yoshida et al. 2013, *Scand J Med Sci Sports* — τ_{PCr} by muscle. PMID 23662804, DOI 10.1111/sms.12081.
- Parolin ML et al. 1999, *Am J Physiol Endocrinol Metab* — glycogen phosphorylase / PDH activation kinetics during maximal intermittent exercise (glycolytic activation over the first ~6 s). PMID 10567017, DOI 10.1152/ajpendo.1999.277.5.E890.
- Bogdanis GC et al. 1996, *J Appl Physiol* — PCr and aerobic contribution during repeated sprints (fresh 30 s sprint $\rightarrow$ PCr $\approx$ 17%; PCr resynthesis correlates with power recovery at $r = 0.84\text{--}0.91$; PCr 78.7% recovered at 3.8 min; the 'near-full by 10 s' figure is *sprint 2*, from 78.7% recovered — mis-cited in v0.6, corrected in v0.7). PMID 8964751, DOI 10.1152/jappl.1996.80.3.876.
- Bogdanis GC et al. 1998, *Acta Physiol Scand* 163(3):261–272 — power output and muscle metabolism after *fresh* 10 s and 20 s maximal sprints, with early (~2 min) recovery of PCr and reproducible power — the fresh-sprint conditions v0.6 lacked (verify exact percentages against the publisher PDF). PMID 9715738, DOI 10.1046/j.1365-201x.1998.00378.x.
- Bogdanis GC et al. 1995, *J Physiol* 482(2):467–480 — PCr resynthesis after a 30 s maximal sprint is *slower* than after longer, milder exercise (the acidotic-domain caution behind §4.1a). PMID 7714837, DOI 10.1113/jphysiol.1995.sp020533.
- Gaitanos GC et al. 1993, *J Appl Physiol* 75(2):712–719 — human muscle metabolism during 10 $\times$ 6 s maximal sprints: glycolytic ATP contribution falls from ~40% (sprint 1) to <10% (sprint 10) as PCr and aerobic metabolism take over (the repeated-sprint partition tested in §6.10). PMID 8226455, DOI 10.1152/jappl.1993.75.2.712.
- *Scientific Reports* 2024 — repeated-sprint ATP partitioning by system from vastus-lateralis biopsy during cycle ergometry (compartment-level, in-modality; §6.10). DOI 10.1038/s41598-024-78916-z.
- González-Alonso J et al. 2000, *J Physiol* — heat production at exercise onset; PCr + glycogenolysis initially provide most energy. PMID 10766936, DOI 10.1111/j.1469-7793.2000.00603.x.
- Bangsbo J et al. 1990, *J Physiol* — anaerobic energy production and O₂ deficit during exhaustive exercise. PMID 2352192, DOI 10.1113/jphysiol.1990.sp018000.

Sourcing note: equations for the incumbent models were cross-checked against PubMed Central full text (PMC7552657, PMC10638188); typeset equations there were image-embedded, so published closed forms were reconstructed from the surviving prose and standard literature. Verify exact coefficients against publisher PDFs before hard-coding.